الطبقة الثانية والمصائب الآتية

Lecture 2 – Layer 2 (Ethernet) Security and Protection

> **Layer 2 (Data Link)** : provides **local** network communication using frames, MAC addresses, switches and VLANs.

> **Ethernet** : the most common wired LAN technology; information is carried in **frames**.

Q/ What are the fields of an Ethernet frame ?

1. **Destination MAC**: the intended receiver.
2. **Source MAC**: the sending interface — switches use it for **learning**.
3. **Type/Length**: the carried protocol (IPv4, IPv6, ARP).
4. **Payload**: the actual information.
5. **FCS (Frame Check Sequence)**: detects transmission errors; damaged frames are discarded.

MAC address	IP address
Layer 2	Layer 3
Local frame delivery	Communication between networks
Used by switches	Used by routers
e.g. 00:25:96:FF:12:34	e.g. 192.168.10.5

A MAC address is **not strong proof of identity**: software can change or imitate it.

How a switch works

> **MAC address table** : maps MAC addresses to switch ports.

The switch **learns from the source MAC** but **forwards by the destination MAC**.

Situation	Switch action
Known destination MAC	Forward only to the correct port

Situation	Switch action
Unknown destination MAC	**Flood** to all other ports in the same VLAN (except the incoming one)
Broadcast FF:FF:FF:FF:FF:FF	Send to all ports in the broadcast domain (except the incoming one)

Q/ Why does Layer 2 need security ?

- Its protocols were built for fast local communication and **trust** what devices send: the switch learns any source MAC, ARP accepts IP-to-MAC claims, DHCP clients accept configuration, STP exchanges BPDUs — all can be abused.

Attacks and protections

> **MAC flooding** : the attacker sends many frames with **fake source MACs** to fill/disturb the MAC table; unknown unicast is then flooded, exposing traffic and reducing performance.

> **MAC spoofing** : the attacker uses **another device's MAC** to imitate it.

	MAC flooding	MAC spoofing
Goal	Disturb the MAC table	Imitate another device
Protection	**Port Security**	**802.1X**, Port Security, stronger authentication

Q/ What can Port Security do ?

1. **Limit the number** of MAC addresses allowed on a port.
2. Use **static**, **dynamically learned** or **sticky** secure MACs.
3. **Drop** unauthorized traffic or **disable** the port on a violation.

> **ARP (Address Resolution Protocol)** : discovers the MAC address associated with an IPv4 address on the local network.

> **ARP spoofing** : sending **false ARP replies** (e.g. "192.168.1.1 is at CC:CC:...:03", the attacker's MAC), because basic ARP doesn't strongly authenticate replies → traffic goes through the attacker (**Man-in-the-Middle**).

Q/ What are the possible results of ARP spoofing ?

1. Traffic interception. 2. Traffic modification. 3. Session hijacking. 4. Denial of service.

Q/ How to protect against ARP spoofing ?

1. **Dynamic ARP Inspection (DAI)**.
2. **DHCP Snooping**.
3. Network segmentation.
4. Encryption of application traffic.
5. Monitoring for unusual or duplicate ARP entries.

DAI validates ARP using trusted IP-to-MAC bindings, commonly learned from the **DHCP Snooping database**.

> **DHCP** : automatically gives clients an IP address, mask, default gateway, DNS server and lease time.

> **DORA** : the normal DHCP process — **D**iscover → **O**ffer → **R**equest → **A**cknowledge.

Attack	What happens	Effect / protection
Rogue DHCP server	An unauthorized server gives false settings (fake gateway, malicious DNS)	Traffic redirected → **DHCP Snooping**
DHCP starvation	Many fake DHCP requests consume the address pool	Users get no IP → DHCP Snooping + **rate limiting**

Q/ What does DHCP Snooping do ?

1. Classifies switch ports as **trusted** or **untrusted**.
2. Blocks DHCP **server** messages from untrusted ports.
3. Builds a **binding database** (IP, MAC, VLAN, port, lease).
4. Supports **DAI** and **IP Source Guard**.

Trusted ports lead toward the real DHCP server; user-facing ports are untrusted.

> **VLAN** : logically separates a switched network into different **broadcast domains** (students, staff, servers…), reducing exposure and allowing different policies.

VLAN hopping type	Idea	Basic protection
Switch spoofing	Attacker pretends to be a switch and negotiates a **trunk**	Configure user ports as **access** ports; disable automatic trunk negotiation
Double tagging	A frame with **two VLAN tags** crosses VLAN boundaries	Use a dedicated unused **native VLAN**; no user devices on it

> **STP (Spanning Tree Protocol)** : prevents switching loops and elects a **root bridge**.

> **STP manipulation** : an unauthorized switch sends **superior BPDUs** to become root → change topology, redirect traffic, or cause instability.

BPDU Guard	Root Guard
Used on host-facing access ports	Used where the root must not appear
Unexpected BPDU → port disabled	Prevents an unexpected switch from becoming root
Protects edge ports	Protects the **root-bridge position**

> **IEEE 802.1X** : authenticates users/devices **before** normal network access.

802.1X component	Role
Supplicant	The user device requesting access
Authenticator	The switch or access point controlling the port
Authentication server	Verifies credentials

Summary tables

Control	Main purpose
Port Security	Limits which MACs can use a port
Disable unused ports	Reduces unauthorized physical access
VLAN segmentation	Separates users/services into zones
DHCP Snooping	Blocks rogue DHCP; builds trusted bindings

Control	Main purpose
DAI	Validates ARP against trusted bindings
IP Source Guard	Filters source IP/MAC on untrusted interfaces
BPDU Guard	Stops STP on user-facing ports
Root Guard	Protects the intended root
IEEE 802.1X	Authenticates before access

Attack	Objective	Protection
MAC flooding	Fill/disturb the MAC table	Port Security
MAC spoofing	Imitate another device	802.1X, Port Security
ARP spoofing	Redirect or intercept traffic	DAI, DHCP Snooping, encryption
Rogue DHCP	Give clients false configuration	DHCP Snooping
DHCP starvation	Exhaust IP addresses	DHCP Snooping, rate limiting
VLAN hopping	Reach another VLAN	Access mode, no negotiation, secure native VLAN
STP manipulation	Change switching topology	BPDU Guard, Root Guard
Unauthorized connection	Gain network access	Disable unused ports, 802.1X

Q/ University case — fix each problem.

1. All users in one VLAN → separate VLANs (students, teachers, admin, servers, guests).
2. Unused ports active → shut them down and put them in an unused VLAN.
3. Unknown devices connect → 802.1X. Too many MACs on a port → Port Security.
4. Rogue DHCP possible → DHCP Snooping. ARP spoofing → DAI.
5. Students connect switches → BPDU Guard. Unexpected root → Root Guard.

> Don't trust a device just because it's connected; a MAC isn't identity; combine controls (**defense in depth**).

Summary of Network Security L2: Layer 2 (Ethernet) Security. Built only from the lecture file; check the original slides for anything you're unsure about.